

What size should my music studio be?

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In Everest, F. Alton, and Ken C Pohlmann, *Master Handbook of Acoustics* (New York: McGraw-Hill, 2009), p. 247, a range of room ratios are listed to achieve optimal modal distributions using the Bolt range and a set of external sources. In other words, this addresses the question: what size should I make my room to achieve the best even distribution of sound throughout?

In order to facilitate these calculations, I have written an open-source program, [acousticsize](#), that will let you calculate room dimensions that fall within the Bolt range using these secondary sources, based on a height input.

So, for example, if you had an internal room height dimension of 2m, you could do:

```
python ./acousticsize.py 2
```

```
[Sepmeyer B]
```

```
[Height]: 2
```

```
[Width]: 2.56
```

```
[Length]: 3.08
```

```
[Sepmeyer C]
```

```
[Height]: 2
```

```
[Width]: 3.2
```

```
[Length]: 4.66
```

```
[Louden D]
```

```
[Height]: 2
```

```
[Width]: 2.8
```

```
[Length]: 3.8
```

```
[Louden F]
```

```
[Height]: 2
```

```
[Width]: 3.0
```

```
[Length]: 5.0
```

```
[Volkmann]
```

```
[Height]: 2
```

```
[Width]: 3.0
```

```
[Length]: 5.0
```

```
[Boner]
```

```
[Height]: 2
```

```
[Width]: 2.52
```

```
[Length]: 3.18
```